

Cloud LearnX

Mathematics

Circle Theorems — Practice Worksheet

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Topic: Circle Theorems

Instructions:

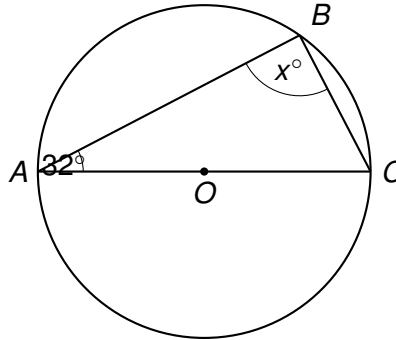
- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

Answer ALL ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1** A , B and C are points on the circumference of a circle with centre O . AC is a diameter of the circle.



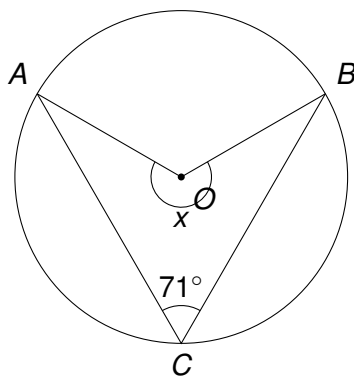
Given that angle $BAC = 32^\circ$, work out the size of angle ABC , marked x° on the diagram.

$$x = \dots\dots\dots^\circ$$

(2)

(Total for Question 1 is 2 marks)

2 A , B and C are points on the circumference of a circle with centre O .



Angle $ACB = 71^\circ$.

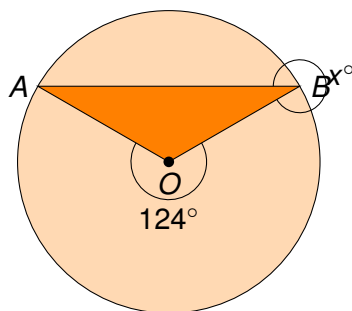
Work out the size of angle AOB , marked x on the diagram.

$x = \dots\dots\dots^\circ$

(2)

(Total for Question 2 is 2 marks)

3 A and B are points on the circumference of a circle with centre O . OA and OB are radii of the circle.



Angle $AOB = 124^\circ$.

Find the value of x .

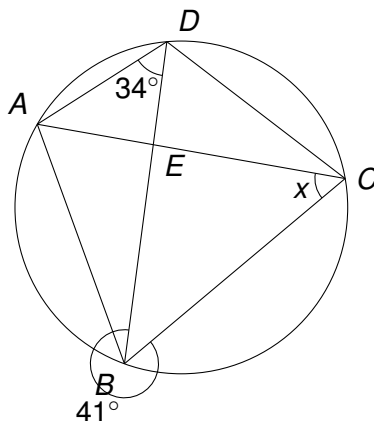
$x = \dots\dots\dots$

(2)

(Total for Question 3 is 2 marks)

- 4 A , B , C and D are points on the circumference of a circle. Chords AC and BD intersect inside the circle at point E .

Angle $ADB = 34^\circ$ and angle $DBC = 41^\circ$.



Angle ADB is the angle subtended by arc AB at D , and angle ACB is the angle subtended by the same arc AB at C .

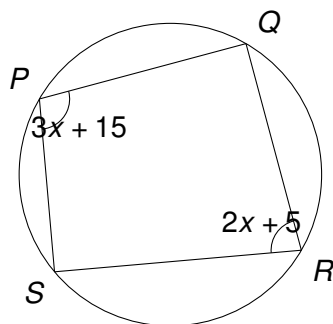
Work out the size of angle x , marked at point C , giving a reason for your answer.

$$x = \dots\dots\dots^\circ$$

(3)

(Total for Question 4 is 3 marks)

- 5 P , Q , R and S are points on the circumference of a circle, forming the cyclic quadrilateral $PQRS$.
Angle $P = (3x + 15)^\circ$ and angle $R = (2x + 5)^\circ$.



Work out the size of angle R .

$$R = \dots\dots\dots^\circ$$

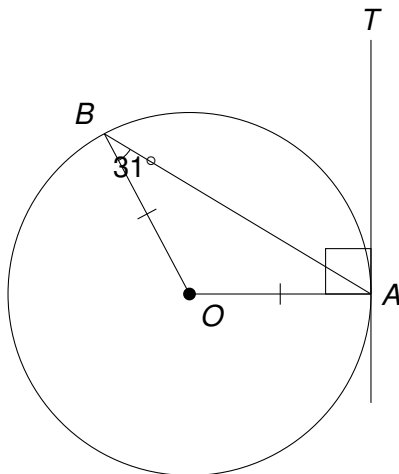
(3)

(Total for Question 5 is 3 marks)

6 A is a point on a circle with centre O . The line TA is a tangent to the circle at A , and B is a point on the circumference such that OB is a radius.

C is the point where the straight line OBT ...

In the diagram below, TA is a tangent to the circle at A , and OAT is a triangle where O is the centre of the circle. OB is a radius drawn to a point B on the circle, and angle $ABO = 31^\circ$.



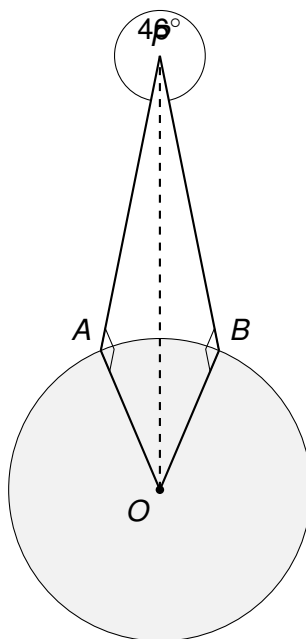
Given that angle $ABO = 31^\circ$, work out the size of angle OAT , and hence find the size of angle BAT .

$BAT = \dots\dots\dots^\circ$
(3)

(Total for Question 6 is 3 marks)

- 7** A satellite dish is held in place by two straight metal support struts, PA and PB , which run from a fixed anchor point P on a pole and touch the circular rim of the dish at points A and B . Each strut is tangent to the circular rim.

O is the centre of the circular rim. The angle between the two struts at P is 46° .



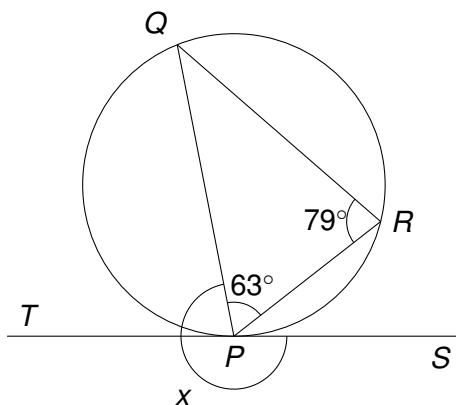
Given that OA and OB are radii of the dish's rim, work out the size of angle AOB .

$$AOB = \dots\dots\dots^\circ$$

(3)

(Total for Question 7 is 3 marks)

- 8 P , Q and R are points on the circumference of a circle. The line TPS is a tangent to the circle at P . Chord PQ and chord PR are drawn, and chord QR is drawn to complete triangle PQR . Angle $QPR = 63^\circ$ and angle $PRQ = 79^\circ$.



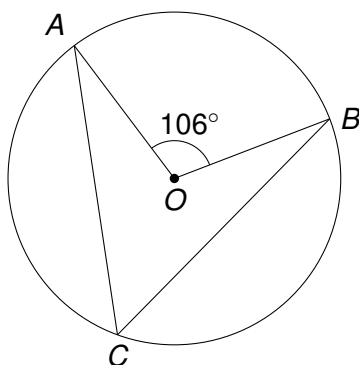
Work out the size of angle x , the angle between the tangent SP and the chord PQ .

$$x = \dots\dots\dots^\circ$$

(3)

(Total for Question 8 is 3 marks)

- 9 A and B are points on the circumference of a circle with centre O . C is also a point on the circumference. OA and OB are radii of the circle, and angle $AOB = 106^\circ$.



- (a) Work out the size of angle OAB .

$$OAB = \dots\dots\dots^\circ$$

(2)

- (b) Work out the size of angle ACB .

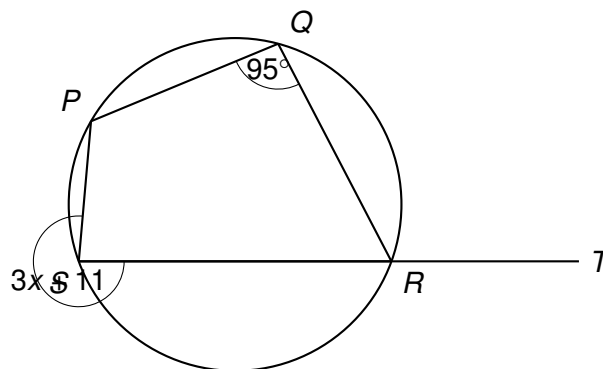
$$ACB = \dots\dots\dots^\circ$$

(2)

(Total for Question 9 is 4 marks)

- 10** P , Q , R and S are points on the circumference of a circle. The side QR is extended to point T , so that SRT is a straight line.

$\angle PQR = 95^\circ$. $\angle PSR = 3x + 11$ (degrees).



Note: $\angle PRT$ is the exterior angle of the cyclic quadrilateral $PQRS$ at vertex R .

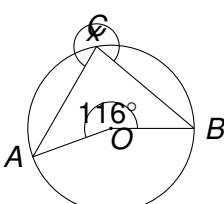
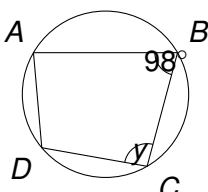
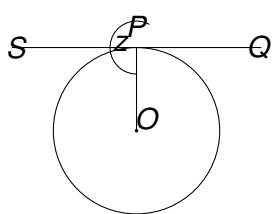
Find the value of x .

$x = \dots\dots\dots$

(3)

(Total for Question 10 is 3 marks)

- 11 For each diagram below, O marks the centre of the circle where shown. Name the circle theorem that lets you find the marked angle, and state the value of that angle.

Diagram	Theorem name and marked angle
	
	
	

Write down the name of the theorem and the value of the marked angle for each of the three diagrams.

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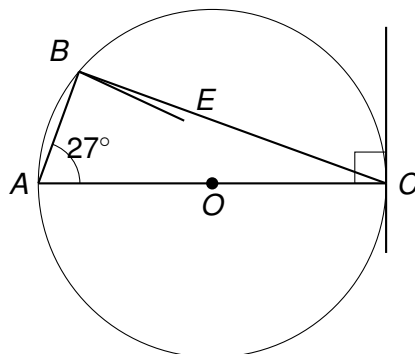
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(3)

(Total for Question 11 is 3 marks)

- 12** A , B and C are points on the circumference of a circle with centre O , where AC is a diameter of the circle. D is also a point on the circumference such that B , O and D are *not* collinear, and BD is a chord. The line AB is extended to meet the tangent at C at the point E , so that angle $ACE = 90^\circ$.



Angle $BAC = 27^\circ$.

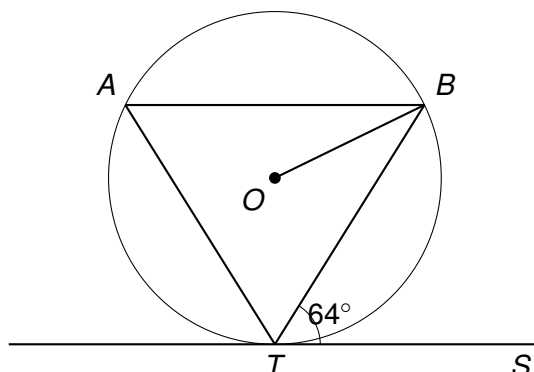
Show that angle $BCE = 27^\circ$, giving a reason for each stage of your working.

(4)

(Total for Question 12 is 4 marks)

- 13** A radio telescope dish has a circular support ring, centre O . A signal cable runs along ST , a tangent to the ring at point T . Two receiver mounts, A and B , lie on the ring, connected to T by struts TA and TB , and to each other by strut AB . The centre O is also connected to mount B by a support rod OB .

The angle between the tangent ST and the strut TB is 64° . Angle $OTA = 90^\circ$ (the rod OT meets the tangent at right angles, but OT is not drawn).



Given that angle $OTA = 90^\circ$, find the size of angle OBA , the angle at mount B between rod OB and strut BA .

$$OBA = \dots\dots\dots^\circ$$

(4)

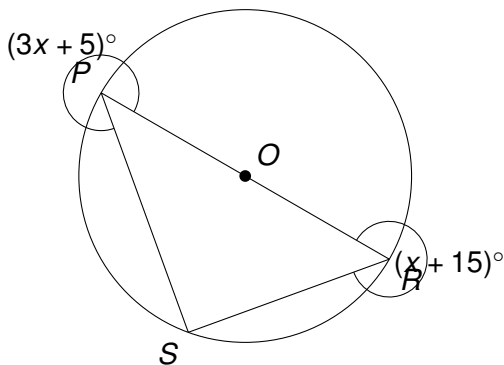
(Total for Question 13 is 4 marks)

- 14** A, B, C and O are points on and inside a circle, where O is the centre. A, B and D lie on the circumference, with AOD a straight line through the centre.

The angle at B in quadrilateral $OABC$...

Points P, Q, R lie on the circumference of a circle with centre O . S is also a point on the circumference, so that $OPSR$ is a cyclic quadrilateral (i.e. O, P, S and R all lie on a circle).

The angle at P is $(3x + 5)^\circ$ and the angle at R is $(x + 15)^\circ$, as shown in the diagram.



- (a) Show that $x = 40$.

(2)

- (b) Work out the size of the angle at the centre, angle POR , reflex.

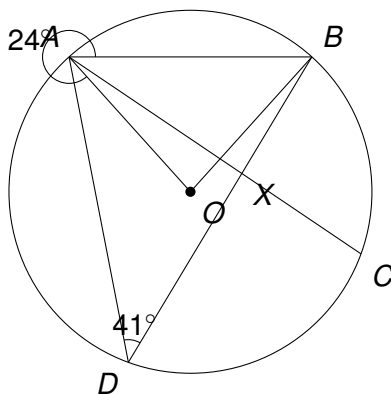
$$POR = \dots\dots\dots^\circ$$

(2)

(Total for Question 14 is 4 marks)

- 15** A , B , C and D are four points on the circumference of a circle with centre O . AC and BD are chords which intersect at point X inside the circle.

$OA = OB$ (both are radii), and angle $OAB = 24^\circ$. Angle $ADB = 41^\circ$.

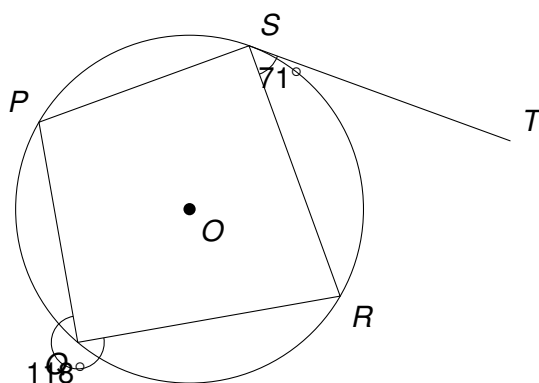


Show that triangle ABX is isosceles.

(4)

(Total for Question 15 is 4 marks)

- 16** Four tracking stations P , Q , R and S observe a satellite that moves along a circular orbit path with centre O . The stations P , Q , R and S all lie on the orbit circle, in that order around the circle. A ground antenna at T lies outside the orbit, and the line TS is a tangent to the orbit circle at S .
Angle $PQR = 118^\circ$. Angle $RST = 71^\circ$.



- (a) Work out the size of angle RSP , giving a reason for your answer.

$$RSP = \dots\dots\dots^\circ$$

(2)

- (b) Hence work out the size of angle PSR 's neighbouring angle PST , using the tangent TS and the alternate segment theorem where appropriate, to find the size of angle SPR .

$$SPR = \dots\dots\dots^\circ$$

(2)

(Total for Question 16 is 4 marks)

TOTAL FOR PAPER IS ?? MARKS